



**CHANDIGARH
UNIVERSITY**

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Academic Session 2026-27

ODD Semester Jul-Dec 2026

INSTITUTE - UIE

Department of Engineering Foundations

Bachelor of Engineering (Computer Science & Engineering)

Subject Name: Logical Thinking & Problem Solving

Code:26CSH-101

Unit No. 1

Chapter No. 1

Lecture No.3

Topic : Hands on session on Flowcharts

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E_Code - E6717

Designation-Assistant Professor

Logical Thinking & Problem Solving



Course Objectives

The course aims to provide a strong foundation in problem-solving using the C programming language.

It enhances students' programming skills by developing logical thinking and algorithmic problem-solving abilities

By mastering C programming concepts, students will be able to design and implement efficient solutions for real-world problems



Learning Outcomes

CO No	Statement
CO1	To remember the concepts related to fundamentals of C language, Algorithm/pseudocode, Flowcharts and Problem solving
CO2	To understand the way of execution and debug programs in C language.
CO3	To apply various constructs, loops, functions to solve mathematical and scientific problem.
CO4	To analyse the dynamic behaviour of memory by the use of pointers.
CO5	To design and develop modular programs for real world problems using control structure and selection structure.



Recap of Previous Lecture

**In this lecture we have
discussed about problem &
problem solving**

**We have discussed about the
phases of problem solving.**

**We have learnt about problem
solving solutions**

CONTENTS

- Introduction to Flowcharts
- Symbols used in flowcharts
- Example Flowcharts



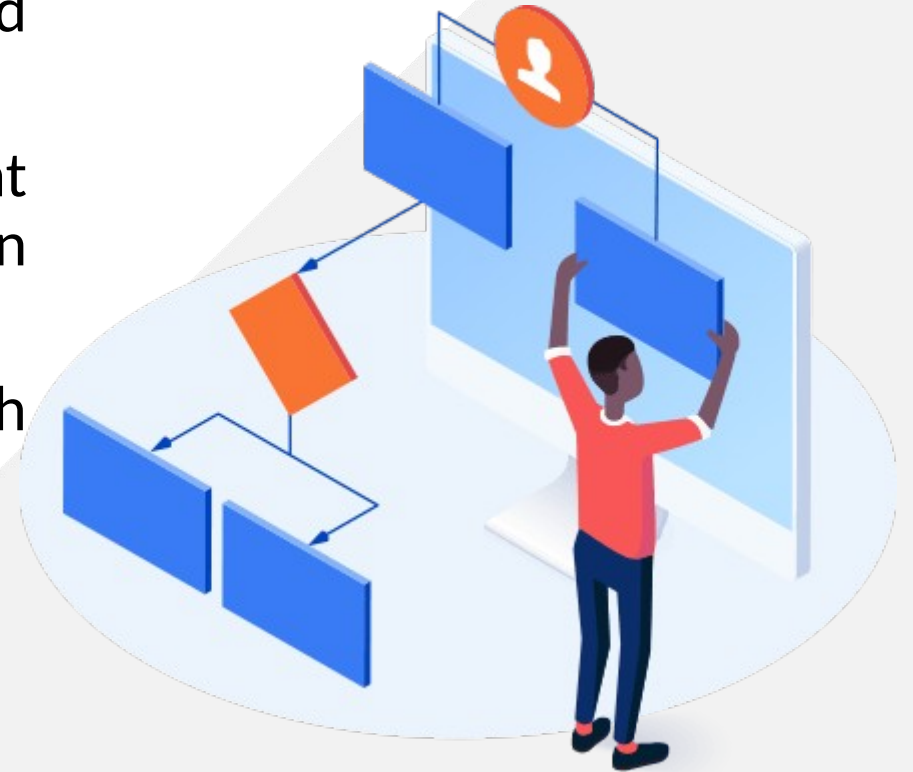
Programming

Uses of Flowchart:

- Planning a new project.
- Documenting a process.
- Modeling a business process.
- Managing workflow.
- Auditing a process.
- Mapping computer algorithms.
- Data management.
- Chemical and process engineering.

FLOWCHART

- A flow chart is a graphical or symbolic representation of a process. Each step in the process is represented by a different symbol and contains a short description of the process step.
- It uses several geometrical figures to represent the operations, and arrows to show the direction of flow.
- The flowchart symbols are linked together with arrows showing the process flow direction.



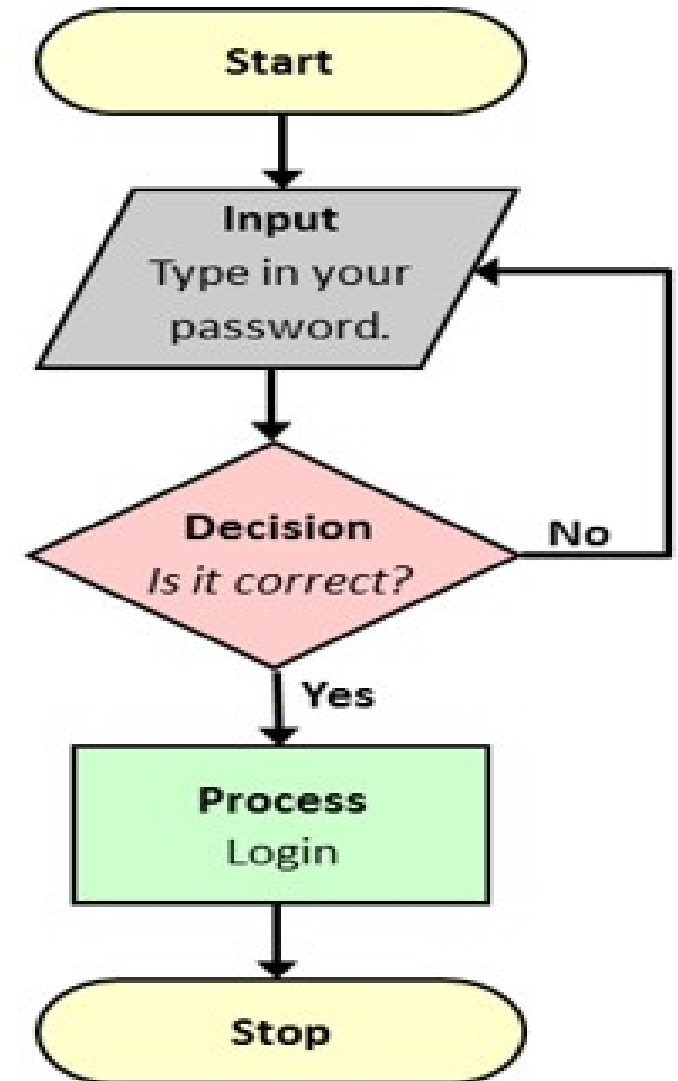
Flowcharts Symbols:

Flow Chart Symbol	Meaning	Explanation
	Start and end	The symbol denoting the beginning and end of the flow chart.
	Step	This symbol shows that the user performs a task. (Note: In many flow charts steps and actions are interchangeable.)
	Decision	This symbol represents a point where a decision is made.
	Action	This symbol means that the user performs an action. (Note: In many flow charts steps and actions are interchangeable.)
	Flow line	A line that connects the various symbols in an ordered way.

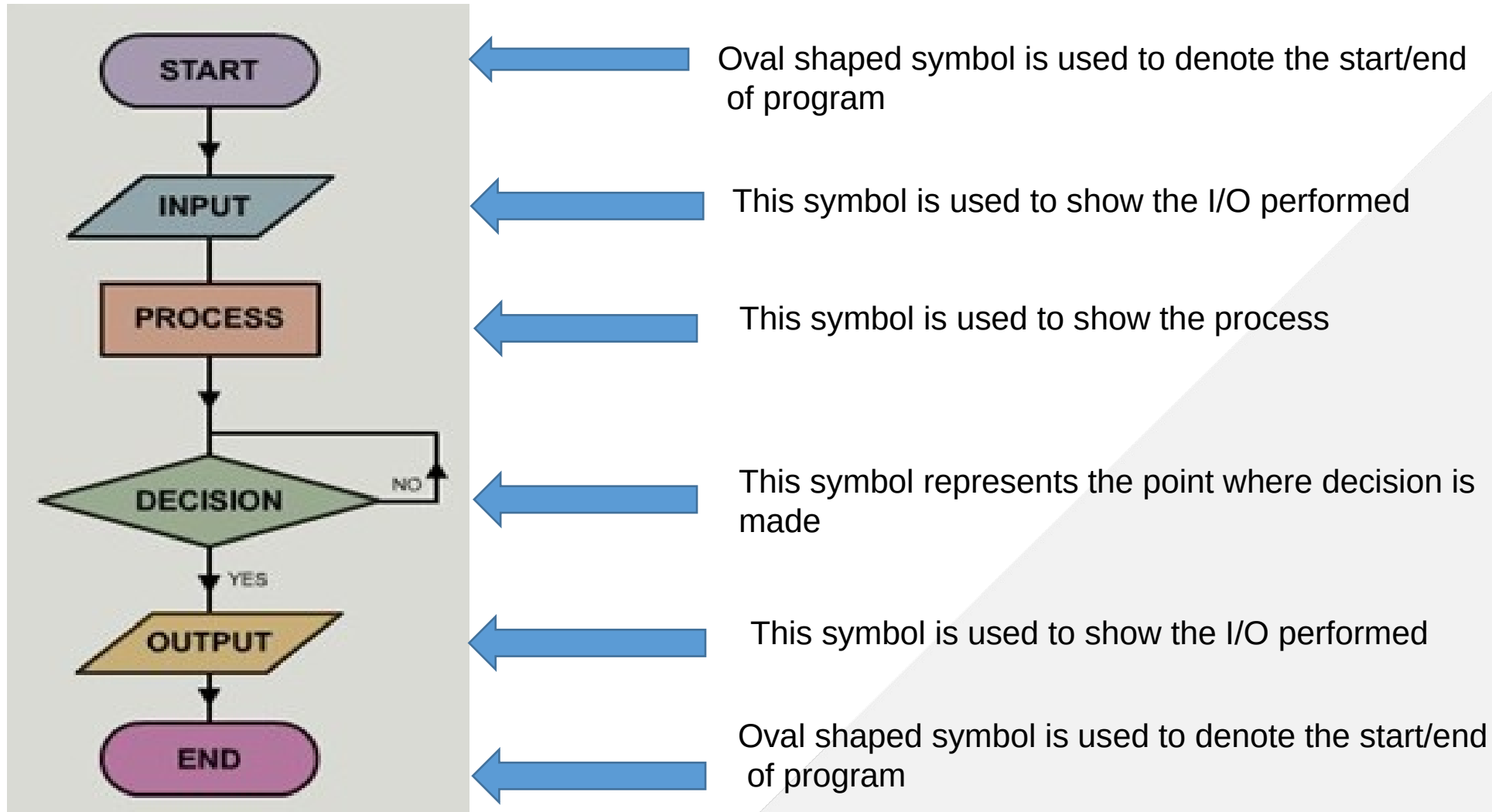
Flowchart – shows steps linked



Computer login program



IDEA OF FLOWCHART



EXAMPLES OF FLOWCHART AND ALGORITHMS

Example: Add 10 and 20

- Start
- Initialize sum = 0 (PROCESS)
- Enter the numbers (I/O)
- Add them and store the result in sum (PROCESS)
- Print sum (I/O)
- Stop

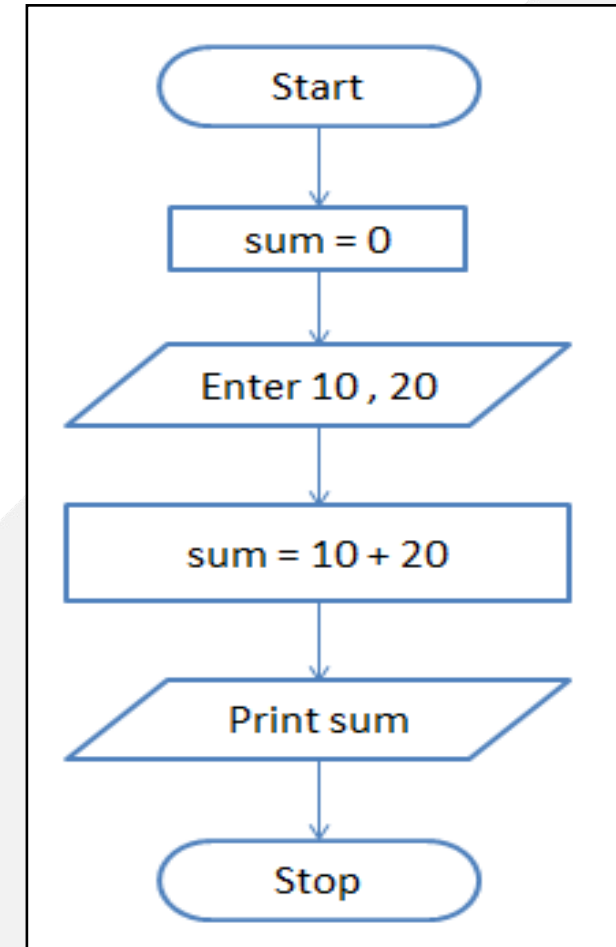


Figure 1.9 flowchart for addition of two numbers

EXAMPLE: CALCULATE INTEREST OF BANK DEPOSIT

Algorithm:

- Step 1: Start
- Step 2: Read amount,
- Step 3: Read years,
- Step 4: Read rate,
- Step 5: Calculate the interest with formula
 $\text{Interest} = \text{Amount} * \text{Years} * \text{Rate} / 100$
- Step 6: Print interest
- Step 7: Stop

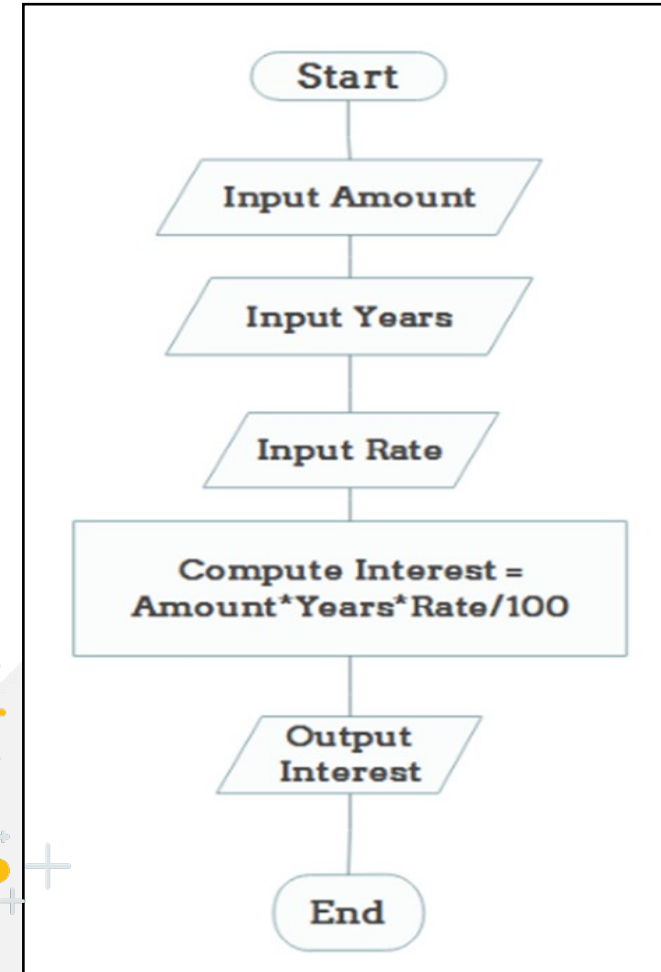
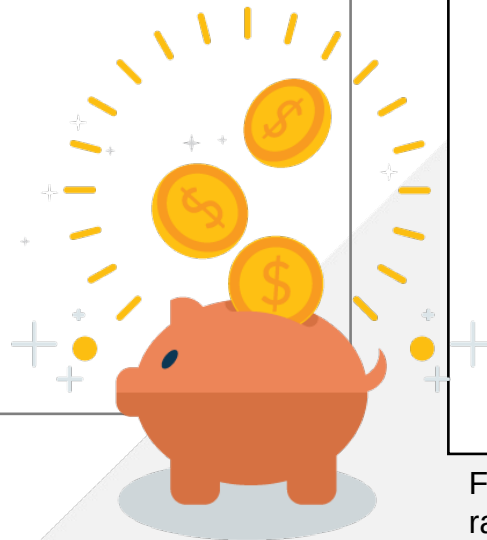


Figure 1.10 flowchart to calculate interest rate

DETERMINE WHETHER THE NUMBER IS EVEN OR ODD

Algorithm:

- Step 1: Start
- Step 2: Read number N,
- Step 3: Set remainder as N modulo 2,
- Step 4: If remainder is equal to 0 then number N is even, else number N is odd.
- Step 5: Print output.
- Step 6: End

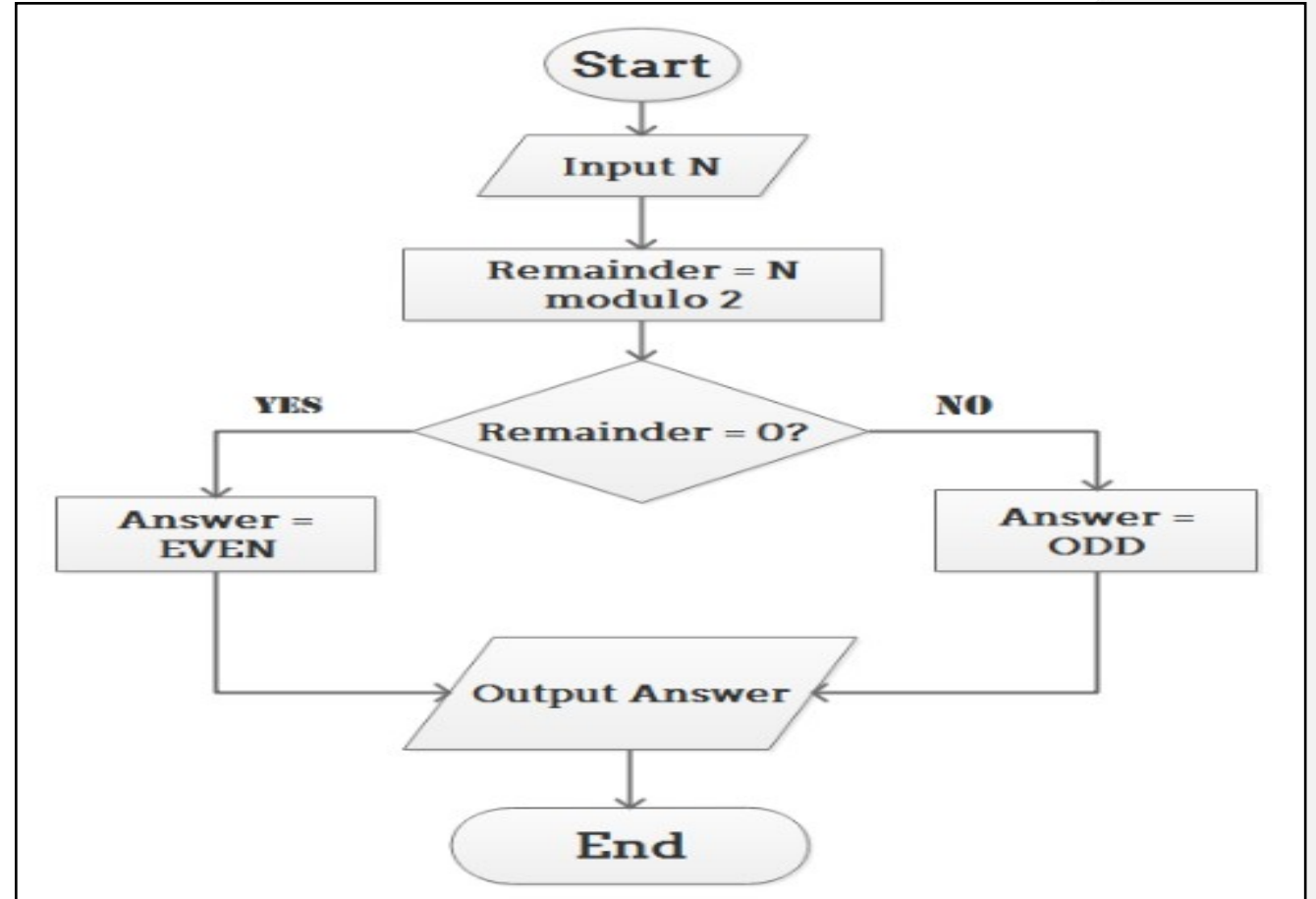


Figure 1.11 flowchart for even/odd

DETERMINE WHETHER TEMPERATURE IS ABOVE OR BELOW FREEZING POINT

Algorithm:

- Step 1: Start
- Step 2: Input temperature,
- Step 3: If it is less than 32,
then print "below freezing point",
otherwise print "above freezing point"
- Step 4: End

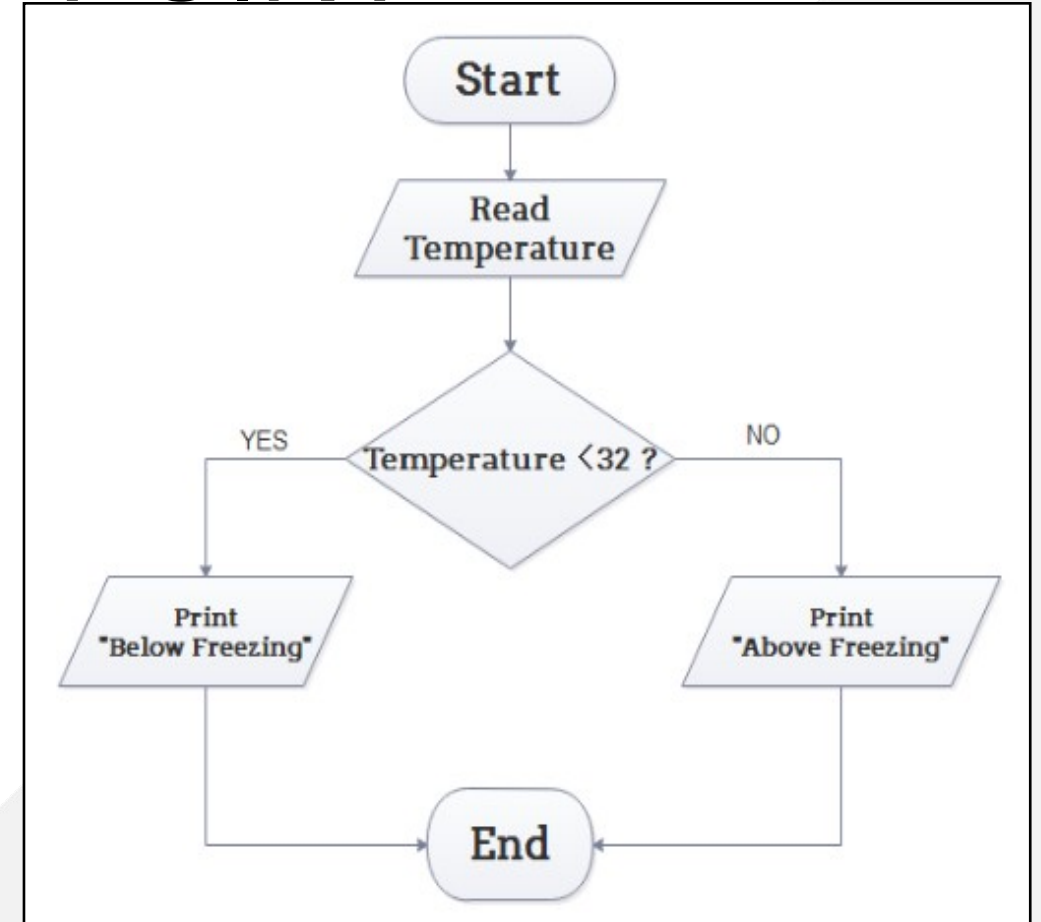


Figure 1.12 flowchart to determine temperature

DETERMINE WHETHER STUDENT PASSED THE EXAM OR NOT

Algorithm:

- Step 1: Start
- Step 2: Input grades of 4 courses M1, M2, M3 and M4,
- Step 3: Calculate the average grade with formula $\text{Grade} = (M1 + M2 + M3 + M4) / 4$
- Step 4: If the average grade is less than 60, print "FAIL", else print "PASS".
- Step 5: End

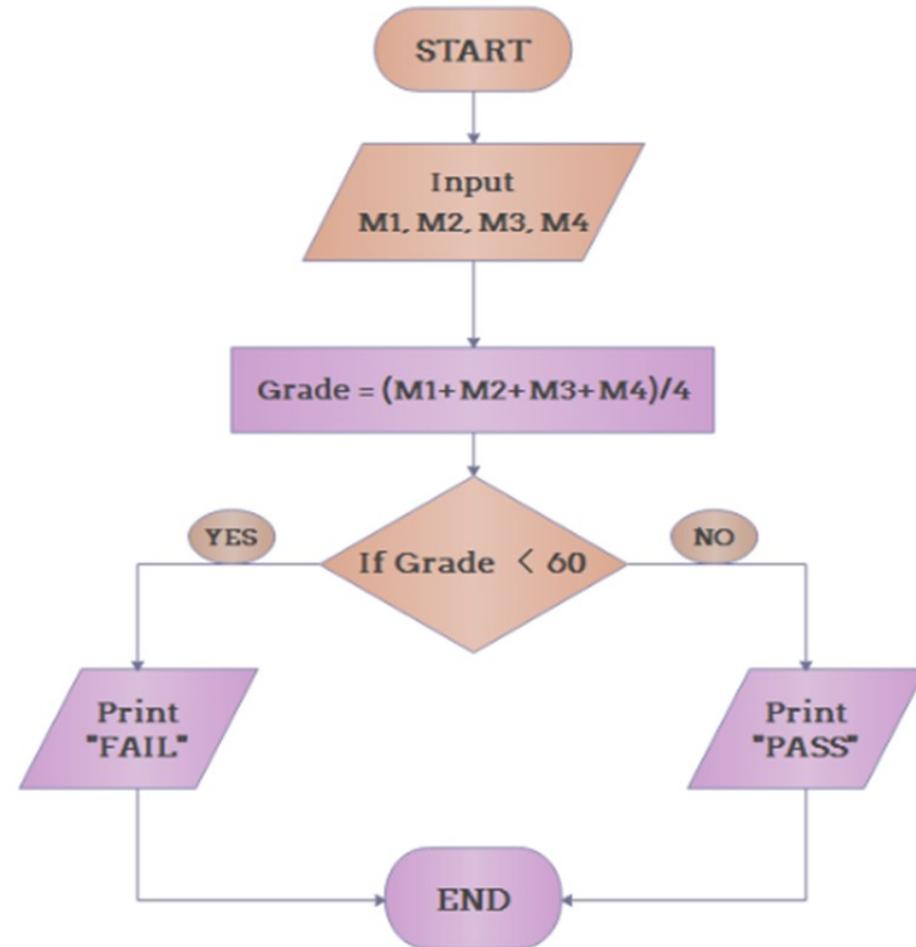


Figure 1.13 flowchart to print pass/fail

Print Hello World 10 times

- Start
- Initialize count = 0 (PROCESS)
- Print Hello World (I/O)
- Increment count by 1 (PROCESS)
- Is count < 10 (DECISION)
- if YES go to step 2
else Stop
- Stop

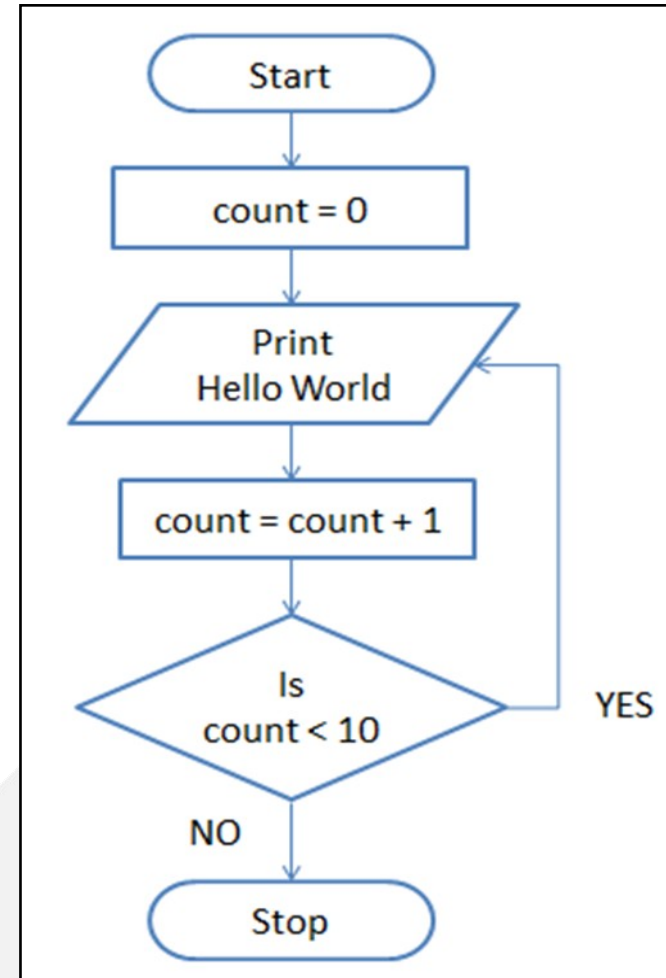


Figure 1.14 flowchart to print hello 10 times

Key Differences Between Algorithm and Flowchart

- An algorithm involves a combination of sequential steps to interpret the logic of the solution. In contrast, a flowchart is the pictorial illustration of the algorithm.
- A flow chart is more understandable as compared to the algorithm.
- The algorithm is written in a language that can be perceived by humans. On the other hand, the flowchart is made up using different shapes and symbols.
- There are no stringent rules are implemented in the algorithms while the flowchart is abode by predefined rules.
- Errors and bugs are easily detected in the algorithm as compared to the flow charts.
- Flow charts are simple to create. On the contrary, the construction of the algorithm is complex.

ADVANTAGES OF FLOWCHART

- It is a convenient method of communication.
- It indicates very clearly just what is being done, where a program has logical complexities.
- A key to correct programming.
- It is an important tool for planning and designing a new system.
- It clearly indicates the role-played at each level.
- It saves the inconveniences in future and serves the purpose of documentation for a system.
- It provides an overview of the system and also demonstrates the relationship between various steps.
- Facilitates troubleshooting.
- It promotes logical accuracy.
- It makes sure that no logical path is left incomplete without any action being taken.

DISADVANTAGES

- The flowchart is a waste of time and slows down the process of software development.
- The flowchart is quite costly to produce and difficult to use and manage.
- Flowcharts are not meant for man to computer communication.
- Sometimes the Complex logic of the program logic is quite complicated to draw out on by using different defined shapes. In that case, flowchart becomes complex and clumsy. This will become a pain for the user, resulting in a waste of time and money trying to correct the problem
- If you need to modify or alternate the process then it will be very hard to do in the flowchart. Because either you will have to erase the end of the flowchart or start.

Summary

In this lecture we have discussed about need and use of flowcharts

In this lecture we have learnt how to design flowchart , how to use specific symbol .

We have also discussed some advantages , disadvantages and applications of Flowcharts

Next Lecture

- An algorithm is a step-by-step procedure used to solve a problem logically and systematically.
- It consists of clearly defined instructions with a proper start and end point.
- Algorithms help in planning program logic before writing actual code.
- A good algorithm should be simple, efficient, accurate, and easy to understand.

Frequently Asked question

Q1: What is a sub-process in a flowchart?

Answer: Sometimes complex processes are broken down into smaller sub-processes for clarity. So a flowchart can point to a different sub process within its flow. The predefined process symbol is used to show such sub-processes.

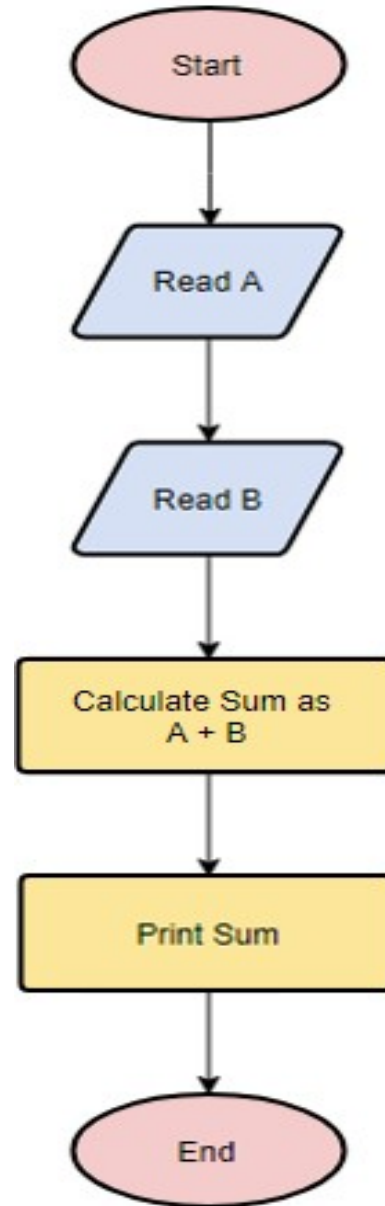
Q2: How are flowcharts used in computer programming?

Answer: A computer program consists many processes and flows. Flowcharts are used to visualize the processes and make it understandable for nontechnical people. They are also used to visualize algorithms and comprehend pseudo code which is used in programming.

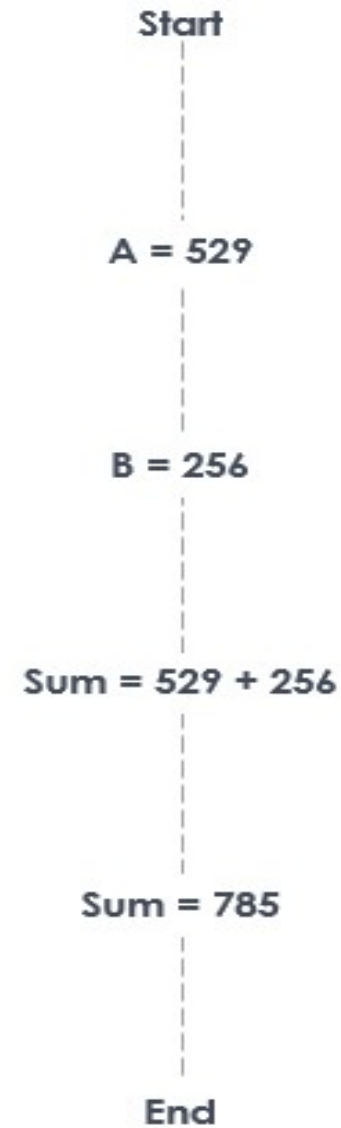
Q3: Which online tools can I use to create flowcharts?

Answer: There are numerous tools which you can find easily. Some of them are: Raptor, Visual Paradigm, Gliffy, and Creately

Q4: Draw the flowchart to represent the sum of two numbers.



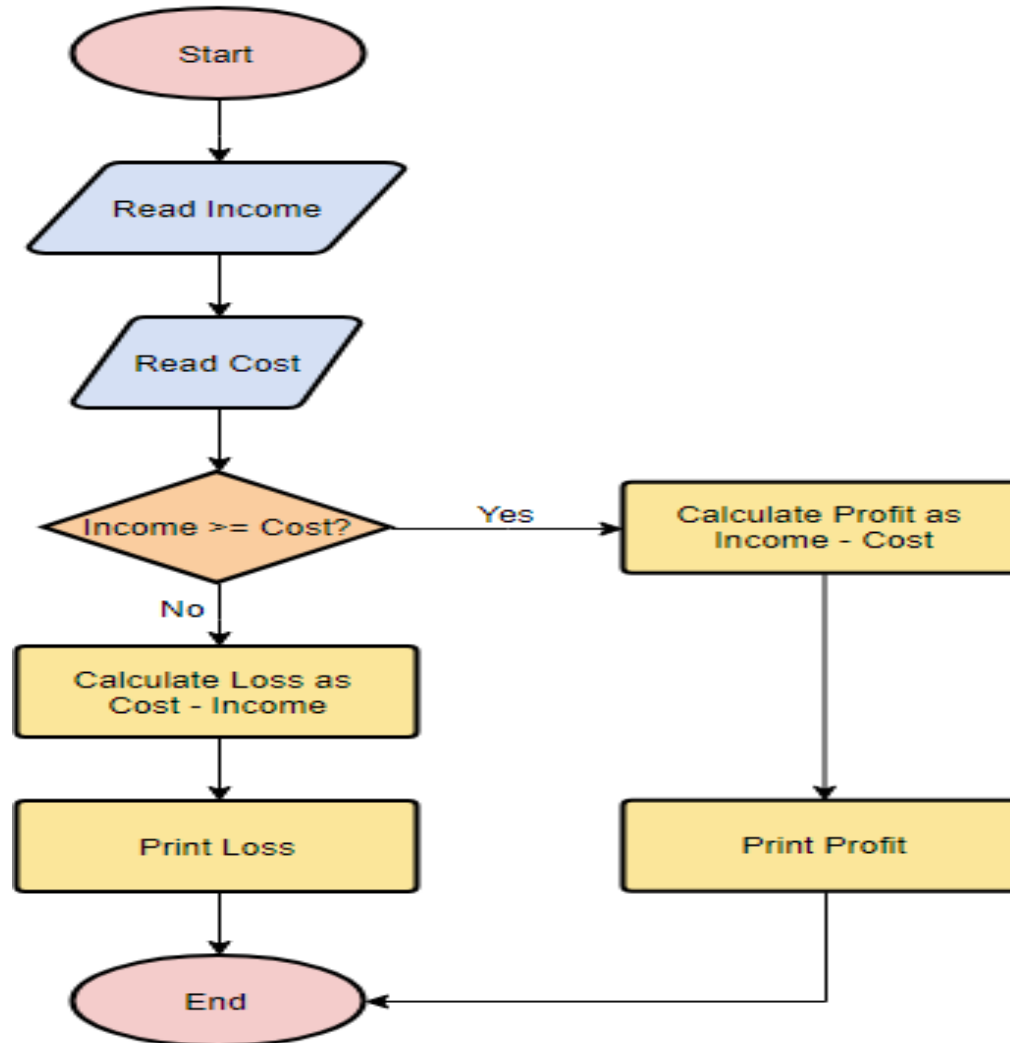
Find the sum of 529 and 256



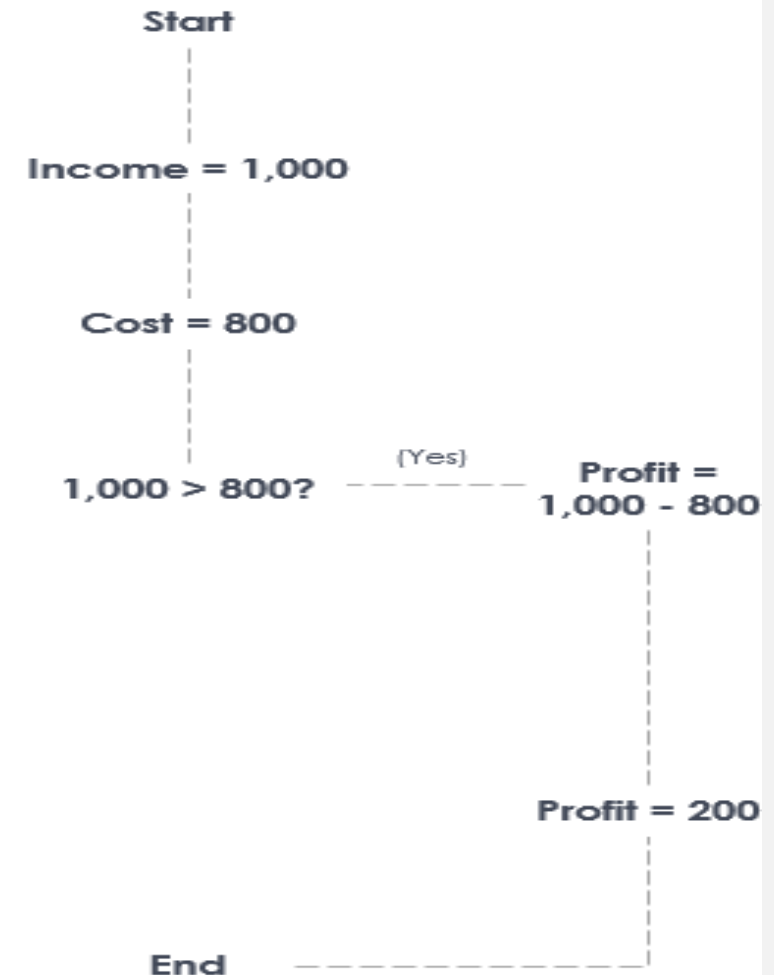
Q5 Draw the flowchart to calculate profit and Loss.

Answer: Flowchart -Calculate Profit and Loss

The flowchart example below shows how profit and loss can be calculated.



Find the profit/loss when
income = 1,000, cost = 800

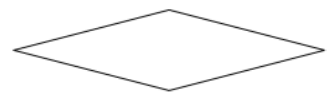


Assessment Questions.

1. The symbol _____ denotes



- a) I/O
- b) Flow
- c) Terminal
- d) Decision

2. The box  denotes?

- a) Decision
- b) Initiation
- c) Initialization
- d) I/O

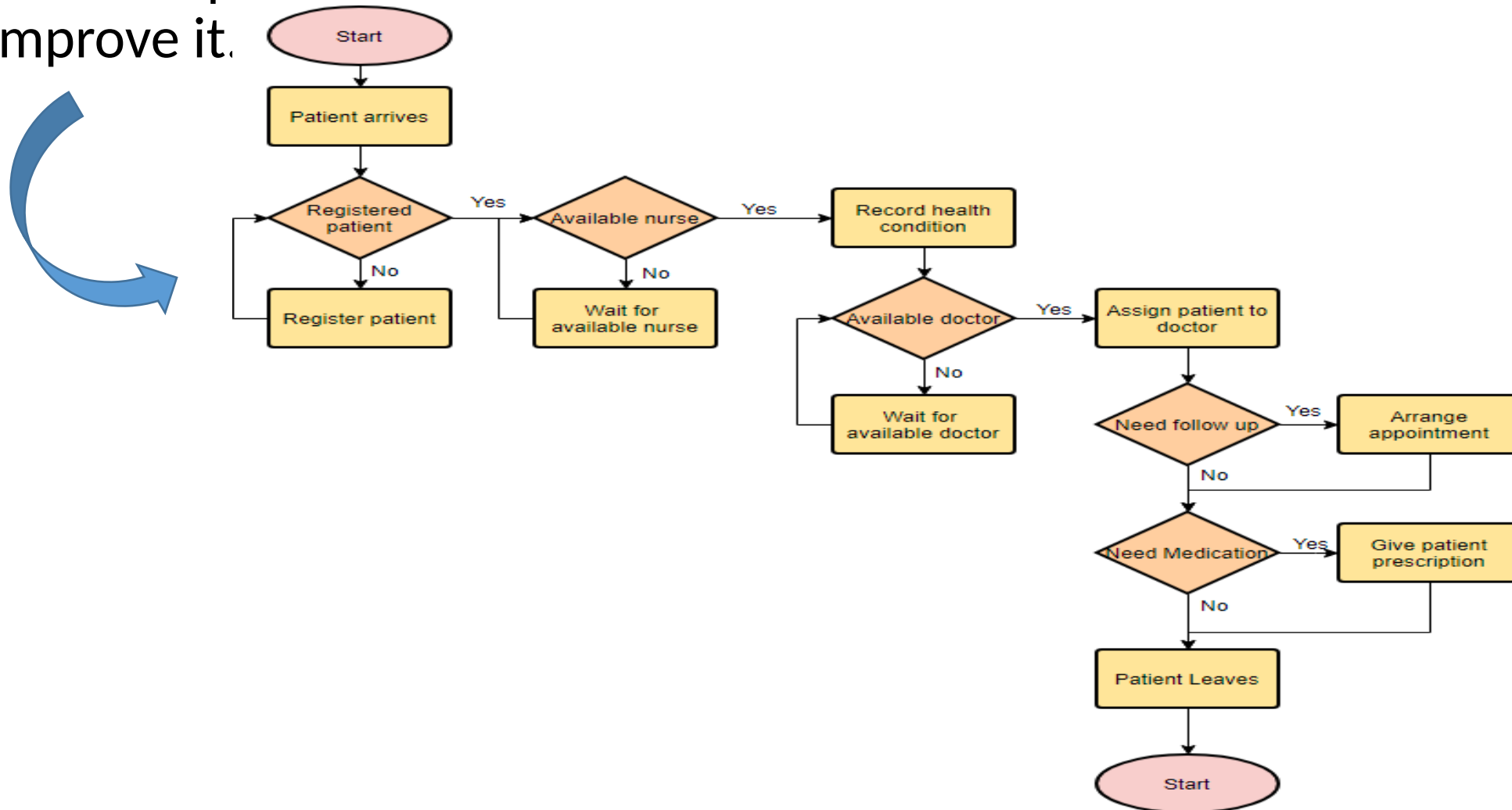
3. Design a flowchart to check leap year

4. Design a flowchart to find sum of n natural numbers

5. Design a flowchart of Quadratic equation

Discussion forum.

Students please have a look at this Medical Service Flowchart and try to improve it.



REFERENCES

Reference Books:

- [1] Programming in C by Reema Thareja.
- [2] Programming in ANSI C by E. Balaguruswamy, Tata McGraw Hill.
- [3] Programming with C (Schaum's Outline Series) by Byron Gottfried Jitender Chhabra, Tata McGraw Hill.
- [4] The C Programming Language by Brian W. Kernighan, Dennis Ritchie Pearson education.

Websites:

<https://medium.com/@warren2lynch/a-comprehensive-guide-for-flowchart-over-examples-785d6dfdc380>

<https://www.geeksforgeeks.org/an-introduction-to-flowcharts/>

<https://www.programiz.com/article/flowchart-programming>

YouTube Links:

Introduction to Creating Flowcharts: <https://youtu.be/SWRDqTx8d4k>

Introduction to Flowchart Symbols: <https://youtu.be/kxZJv56BxU8>



NPTEL Lectures (Free)

1. Problem Solving Through Programming in C

[NPTEL-SWAYAM Course Portal](#)

2. Programming in C – NPTEL

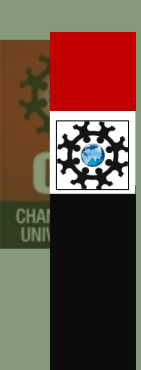
https://onlinecourses.nptel.ac.in/noc26_cs01/preview

Coursera Courses

1. **Programming for Everybody (Getting Started with C) – Beginner Friendly**
<https://www.coursera.org/learn/c-for-everyone>
2. **Duke University – Introductory C Programming Specialization**
<https://www.coursera.org/specializations/c-programming>
3. **Coursera – Beginner C Programming Courses**
<https://www.coursera.org/courses?query=beginner%20c%20programming>

Class-wise feedback





THANK YOU

